

First Consequences of the ZFC Axioms

Extensionality, Comprehension, Pairing, Union — classes, the empty set, pairs,
and the first numbers

Jan Reimann

We take the axioms of [ZFC](#) as given and start deriving things — beginning with Axioms **1**, **3**, **4**, **5** (Extensionality, Comprehension, Pairing, Union).

Extensionality and Comprehension

Axiom 1 (Extensionality). A set is *completely determined by its elements*:

$$\forall z (z \in x \leftrightarrow z \in y) \rightarrow x = y.$$

This does more than it seems. It also fixes what our universe contains. If we allowed an object with no members that is *not* the empty set — a duck D and a badger B , say — then $\forall z (z \in B \leftrightarrow z \in D)$ would hold vacuously, forcing $B = D = \emptyset$. So every object in our universe is a set, and hereditarily so: its members, their members, and so on are all sets. The universe of hereditary sets is informally called V .

Axiom 3 (Comprehension / Separation). For each formula φ (with y not free),

$$\forall v \exists y \forall x (x \in y \leftrightarrow x \in v \wedge \varphi(x)).$$

Subsets are *separated out of a set that already exists*. This restriction is the whole point, as the next section shows.

How Comprehension avoids Russell's paradox

Naive comprehension — “for every property Q , the set $\{x : Q(x)\}$ exists” — is inconsistent. With $Q(x) \equiv x \notin x$ we get Russell's set R and the contradiction $R \in R \leftrightarrow R \notin R$.

Under Axiom 3 we may only form, relative to a set v ,

$$R_v = \{x \in v : x \notin x\}.$$

Run Russell's argument on this. If $R_v \in v$, then $R_v \in R_v \leftrightarrow R_v \notin R_v$ — still a contradiction. The escape is that we are no longer forced into it: we simply conclude

$$R_v \notin v.$$

Since v was arbitrary, every set omits some object, and there is no set of all sets:

Theorem (no universal set). $\forall v \exists R (R \notin v)$.

Whether this is a *desirable* feature is a fair question — a foundation in which “the collection of all sets” is not itself an object can feel like a defect. It is better read as the price of consistency: the same restriction that blocks Russell blocks the universal set.

Formal definitions versus their metatheoretic meaning

A defined symbol is introduced by a **purely symbolic** clause. For intersection:

$$\text{int}(a, b, y) :\iff \forall x (x \in y \leftrightarrow (x \in a \wedge x \in b)).$$

Informally we read $y = a \cap b$, but that reading lives in the metatheory. Before we are entitled to write the *term* $a \cap b$, we must check that it names exactly one set:

Lemma. $\forall a, b \exists! y \text{ int}(a, b, y)$.

Existence is Comprehension applied to $v = a$ with $\varphi(x) \equiv x \in a \wedge x \in b$. *Uniqueness* is Extensionality: if $\text{int}(a, b, y)$ and $\text{int}(a, b, z)$, then $\forall x (x \in y \leftrightarrow x \in z)$, hence $y = z$.

This is the standard pattern. We name an operation (or a constant) only after proving that the defining formula is satisfied by **exactly one** set — just as one proves $\sqrt{2}$ is the unique $y > 0$ with $y^2 = 2$ before using the notation.

Classes and the class-builder notation

Let $\varphi(x)$ be any formula. Informally $\{x : \varphi(x)\}$ is a **class** — this is just notation.

- **If** there is a set A with $\forall x (x \in A \leftrightarrow \varphi(x))$, then A is unique by Extensionality; we write $A = \{x : \varphi(x)\}$ and say the class **exists** (is a set).
- **Otherwise** $\{x : \varphi(x)\}$ is a **proper class** — a figure of speech, not an object.

Two examples:

$$\varphi(x) \equiv x \notin x \quad \rightsquigarrow \quad \text{ZFC} \vdash \neg \exists y \forall x (x \in y \leftrightarrow x \notin x),$$

so Russell's class is proper;

$$\varphi(x) \equiv x = x \quad \rightsquigarrow \quad \neg \exists A \forall x (x \in A),$$

so the **universal class** $V = \{x : x = x\}$ is proper. Saying “ V does not exist” is just another way of writing $\neg \exists A \forall x (x \in A)$; there is no more of a puzzle here than in “trolls do not exist.”

Finally, $\{x \in z : \varphi(x)\}$ abbreviates $\{x : x \in z \wedge \varphi(x)\}$, and Comprehension says **these always exist**.

Building sets from nothing

The empty set

$\emptyset$ the unique y with $\forall z (z \notin y)$.

Existence: take any set (Axiom 0) and apply Comprehension with an always-false φ , e.g. $x \neq x$.

Uniqueness: Extensionality. At this point $\emptyset$ is the **only specific** set we have proved to exist.

Pairs

Pairing gives $\exists z (x \in z \wedge y \in z)$; the z may have other elements, so trim it with Comprehension to $\{w \in z : w = x \vee w = y\}$. This licenses:

$$\{x, y\} := \{w : w = x \vee w = y\}$$

$$\{x\} := \{x, x\}$$

$$\langle x, y \rangle = (x, y) := \{\{x\}, \{x, y\}\}$$

The one property the ordered pair must have is

$$\langle x, y \rangle = \langle x', y' \rangle \rightarrow x = x' \wedge y = y'$$

(prove it by splitting into the cases $x = y$ and $x \neq y$). Any other definition with this property would serve equally well; the bracket notation $\langle \cdot, \cdot \rangle$ marks that we mean *this* one.

We can now start counting:

$$0 = \emptyset, \quad 1 = \{0\} = \{\emptyset\}, \quad 2 = \{0, 1\} = \{\emptyset, \{\emptyset\}\}.$$

The axioms so far generate infinitely many different sets — for example, $\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \dots$ — but **none with more than two elements**.

Union: getting bigger sets

Axiom 5 — Union $\forall \mathcal{F} \exists A \forall Y \forall x (x \in Y \wedge Y \in \mathcal{F} \rightarrow x \in A)$

$\bigcup \mathcal{F} := \{x : \exists Y \in \mathcal{F} (x \in Y)\}$ — the members of members of $\mathcal{F}$.

It exists: take A from the axiom and separate $\{x \in A : \exists Y \in \mathcal{F} (x \in Y)\}$.

$$u \cup v := \bigcup \{u, v\}$$

The axiom is stated for arbitrary families so that it covers **infinite** unions too, not just $u \cup v$.

Worked example: the number 3. We have 0, 1, 2. Then:

1. $\{0, 1\}$ by Pairing;
2. $\{2\} = \{2, 2\}$ by Pairing;
3. $\mathcal{F} = \{\{0, 1\}, \{2\}\}$ by Pairing;
4. $\bigcup \mathcal{F} = \{0, 1, 2\} =: 3$.

Equivalently $3 = 2 \cup \{2\}$. In general the **successor** is

$$S(x) = x \cup \{x\}, \quad 3 = S(2), \quad 4 = S(3), \quad \dots$$